

Biotechnology and Food Engineering

Process Biotechnology

Spring 2026

Tutorial #3 – Growth and production, productivity, Monod model and oxygen uptake rate (OUR) – Solutions

Question 1

A factory produces specific protein secreted from a certain yeast strain, in a production 2.5 m³ batch fermenter. The maximal growth rate of this yeast is 0.55 per hour and Monod constant (K_s) of the sugar used in this process is 800 mg/l.

The productivity (volumetric production rate) in the production fermenter is 0.27 gP/l·hr, and the specific production rate of this protein is 0.127 gP/gX·hr. The maximal sugar allowance in the production fermenter is 4% and it is not possible to exceed this concentration, otherwise the desired protein will not be produced. The generation time in the production fermenter is 30% bigger than the minimal generation time of the yeast.

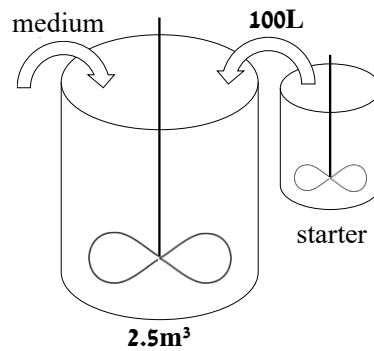
For the initiation of fermentation in the production fermenter, a starter is cultured in a small batch fermenter, with a volume of 100 liter, contains the same medium as in the production fermenter. The fermentation in the starter fermenter is stopped when the yeast concentration reaches to 6 gr/liter. The growth rate at that time point is 0.45 per hour. Inoculation of the production fermenter is done by transferring the whole starter fermenter into the production fermenter, so that the logarithmic growth stage starts immediately and lasts for 10 hours. When the logarithmic stage ends, the fermenter content is harvested and transferred to separation processes.

- A. What is the sugar weight required to be initially added to the production fermenter before inoculation (in kg)?** *(99.64 kg)*
- B. What is the time period of cleaning and preparation in each fermentation cycle of the production fermenter (hr)?** *(8 hr)*

Solution:

Given:

<u>Yeast</u>	<u>Inoculation fermenter</u>	<u>Production fermenter</u>
$\mu_{max} = 0.55 \frac{1}{hr}$	$V_{inc} = 100L$	$V_p = 2.5m^3 = 2500L$
$K_S = 800 \frac{mgS}{L} = 0.8 \frac{gS}{L}$	$X_{finc} = 6 \frac{gX}{L}$	$t_d = 1.3t_{dmin}$
	$\mu_{finc} = 0.45 \frac{1}{hr}$	$Pr = 0.27 \frac{gP}{L \times hr}$
		$t_{log} = 10hr$
		$S_{0max} = 4\% = 40 \frac{gS}{L}$
		$q_P = 0.127 \frac{gP}{gX \times hr}$

**A. $W_{sop'}$ =?**

We need to find the maximal sugar weight that can be added after the starter addition:

$$W_{S_{op}} = S_{0max} \times V_p = 40 \frac{gS}{L} \times 2500L = 100,000gS = 100 KgS$$

The maximal sugar weight include both the sugar comes in the starter and the sugar added during the medium preparation:

$$W_{S_{op}} = W_{S_{op}'} + W_{S_{finc}}$$

We need to find the sugar weight that originates from the starter:

$$W_{S_{finc}} = S_{finc} \times V_{inc}$$

The final sugar concentration at the end of the starter growth is needed. We can find it by the Monod equation, while we have the Monod constant and the growth rate at the end of the growth in the inoculation fermenter:

$$\begin{aligned}\mu &= \mu_{max} \times \frac{S}{K_S + S} \\ 0.45 \frac{1}{hr} &= 0.55 \frac{1}{hr} \times \frac{S_{finc}}{0.8 \frac{gS}{L} + S_{finc}} \\ 0.818 \times \left(0.8 \frac{gS}{L} + S_{finc} \right) &= S_{finc} \\ S_{finc} &= 3.6 \frac{gS}{L}\end{aligned}$$

Therefore, the sugar weight originates from the starter that being added to the fermenter:

$$W_{S_{finc}} = S_{finc} \times V_{inc} = 3.6 \frac{gS}{L} \times 100L = 360gS = 0.36 KgS$$

From here, the sugar weight that should be added to the production fermenter before the addition of the starter:

$$W_{S_{0p}'} = W_{S_{0p}} - W_{S_{finc}} = (100 - 0.36)KgS$$

$$W_{S_{0p}'} = 99.64 KgS$$

B. $t_{maintanance} = ?$

The total time of the fermentation consists of the growth period and the maintenance time:

$$t_{Total} = t_{maintanance} + t_{growth} \rightarrow t_{maintanance} = t_{Total} - t_{growth}$$

The total time of a fermentation cycle can be found from the process productivity:

$$Pr = \frac{\Delta P}{t_T} \rightarrow t_T = \frac{\Delta P}{Pr}$$

We need ΔP that can be calculated from the production yield:

$$Y_{\frac{P}{X}} = \frac{\Delta P}{\Delta X} \rightarrow \Delta P = Y_{\frac{P}{X}} \times (X_f - X_0)$$

We can calculate the production yield from the specific production constant of the yeast:

$$q_P = Y_{\frac{P}{X}} \times \mu \rightarrow Y_{\frac{P}{X}} = \frac{q_P}{\mu}$$

We need the growth rate which can be calculated from the doubling time:

$$\mu = \frac{\ln 2}{t_d}$$

The minimal doubling time is obtained when the growth rate is maximal, and the correlation between the doubling time and the growth rate is:

$$\mu = \frac{\ln 2}{t_d}$$

$$t_{d_{min}} = \frac{\ln 2}{\mu_{max}}$$

Given that $t_d = 1.3t_{d_{min}}$ so by placing in the equation we get that the growth rate in the fermentation:

$$\mu = \frac{\ln 2}{1.3 \times t_{d_{min}}} = \frac{\ln 2}{1.3 \times \frac{\ln 2}{\mu_{max}}}$$

$$\mu = \frac{\mu_{max}}{1.3}$$

$$\mu = \frac{0.55 \frac{1}{hr}}{1.3}$$

$$\mu = 0.423 \frac{1}{hr}$$

Now we can calculate the production yield:

$$Y_{\frac{P}{X}} = \frac{q_P}{\mu} = \frac{0.127 \frac{gP}{gX \times hr}}{0.423 \frac{1}{hr}} = 0.3 \frac{gP}{gX}$$

We can calculate X_0 by using the parameters of the inoculation and production fermenters:

$$X_0 = \frac{X_{f_{inc}} \times V_{inc}}{V_p} = \frac{6 \frac{gX}{L} \times 100L}{2500L} = 0.24 \frac{gX}{L}$$

The biomass concentration at the end of the growth, X_f , we can find from the logarithmic growth equation:

$$\ln \frac{X_f}{X_0} = \mu \times \Delta t \rightarrow X_f = X_0 \times e^{\mu \times \Delta t}$$

The growth period is given in the question - $t_{log} = 10hr$, so:

$$X_f = 0.24 \frac{gX}{L} \times e^{0.423 \frac{1}{hr} \times 10hr} = 16.5 \frac{gX}{L}$$

We need to go back to the production yield to find ΔP :

$$\Delta P = Y_{\frac{P}{X}} \times (X_f - X_0) = 0.3 \frac{gP}{gX} \times (16.5 - 0.24) \frac{gX}{L} = 4.876 \frac{gP}{L}$$

We need to go back to the productivity to find the fermentation cycle time:

$$t_{Total} = \frac{\Delta P}{Pr} = \frac{4.876 \frac{gP}{L}}{0.27 \frac{gP}{L \times hr}} = 18hr$$

From here, the time it takes to clean and prepare the production fermenter is:

$$t_{maintanance} = t_{Total} - t_{growth} = (18 - 10)hr$$

$$t_{maintanance} = 8hr$$

Question 2

Bacteria with the empiric formula of $C_{1.05}H_{1.83}N_{0.32}O_{0.61}S_{0.01}$ is growing on a basic medium containing glycerol ($C_3H_8O_3$) as the carbon source, $(NH_4)_2SO_4$ as the nitrogen source and a trace elements solution as required for growth. The ash percentage in this bacterium is 8% and the growth rate is 0.476 per hour. The fermentation product is a growth hormone, but the amount produced is minor and negligible relative to the amount of biomass produced. It is known that $Y_{p/x}$ equals to $2.5 \frac{mg \text{ Hormone}}{gr \text{ Biomass}}$.

The fermenter is supplied with oxygen enriched air in a flow rate of 100 LPH. The air consists of 42% oxygen and it is supplied in a temperature of 22°C and pressure of 1.5 atmospheres.

Additional data: the fermenter volume is 1 liter and the yield coefficient ($Y_{x/s}$) is $0.6 \frac{gr \text{ Bacteria}}{gr \text{ Glycerol}}$.

What should be the minimal solubility of the air in the fermenter (in percentages) to reach the final concentration of $20 \frac{gr \text{ Biomass}}{liter}$ (12%)

Solution:

Given:

<u>Fermenter</u>	<u>Air</u>	<u>Bacteria</u>
$V_{ferm} = 1L$	$Q_{air} = 100 \frac{L_{air}}{hr}$	$C_{1.05}H_{1.83}N_{0.32}O_{0.61}S_{0.01}$
$S = C_3H_8O_3$	$O_{2in} = 0.42 \frac{L_{O_2}}{L_{air}}$	$Ash = 8\%$
$N = (NH_4)_2SO_4$	$T = 22^\circ C = 295^\circ K$	$\mu = 0.467 \frac{1}{hr}$
	$P = 1.5 atm$	$Y_{\frac{P}{X}} = 2.5 \frac{mgP}{gX}$
		$Y_{\frac{X}{S}} = 0.6 \frac{gX}{gS}$
		$X_f = 20 \frac{gX}{L}$

The minimal solubility of the air will provide the maximal oxygen requirement of the bacteria:

$$\text{minimal solubility}\% = \frac{OUR_{max}}{\left(\frac{Q_{O_2}}{V_{ferm}}\right)_{in}} \times 100\%$$

In this fermentation there is only one microorganism, we know the growth rate and the carbon source is single so the maximal oxygen uptake rate will be at the end of the growth:

$$OUR_{max} = OUR_f = \mu \times X_f \times Y_{\frac{O}{X}} = 0.467 \frac{1}{hr} \times 20 \frac{gX}{L} \times Y_{\frac{O}{X}}$$

We need $Y_{\frac{O}{X}}$ that can be calculated from the oxygen consumption stoichiometry since:

- ✓ The microorganism composition is known - $C_{1.05}H_{1.83}N_{0.32}O_{0.61}S_{0.01} + 8\% Ash$
- ✓ The carbon source is defined - $C_3H_8O_3$
- ✓ The nitrogen source is ammonia salt - $(NH_4)_2SO_4$
- ✓ The growth is for biomass production only - because the product amount is insignificant compared with the biomass.

$$Y_{\frac{O}{X}} = \frac{32 \cdot C + 8 \cdot H - 16 \cdot O}{Y_{\frac{X}{S}} \cdot Mw_S} + 0.01 \cdot O' - 0.0267 \cdot C' + 0.01714 \cdot N' - 0.08 \cdot H'$$

The substrate parameters:

$$Mw_S = a \cdot Mw_C + b \cdot Mw_H + c \cdot Mw_O$$

$$Mw_S = 3 \cdot 12 + 8 \cdot 1 + 3 \cdot 16 = 92 \frac{gS}{moleS}$$

$$C = 3 \quad H = 8 \quad O = 3$$

The bacteria parameters should be calculated:

$$Mw_X = \frac{\alpha \cdot Mw_C + \beta \cdot Mw_H + \gamma \cdot Mw_O + \delta \cdot Mw_N + \varepsilon \cdot Mw_S}{\frac{(100 - Ash)\%}{100\%}}$$

$$Mw_X = \frac{1.05 \cdot 12 + 1.83 \cdot 1 + 0.61 \cdot 16 + 0.32 \cdot 14 + 0.01 \cdot 32}{\frac{(100 - 8)\%}{100\%}}$$

$$Mw_X = 31.51 \frac{gX}{moleX}$$

$$O' = \frac{\gamma \cdot Mw_O}{M \cdot Mw_X} \cdot 100\% = \frac{16 \frac{gO}{moleO} \cdot 0.61 moleO}{31.51 \frac{gX}{moleX} \cdot 1 moleX} \cdot 100\% = 30.97\%$$

$$C' = \frac{\alpha \cdot Mw_C}{M \cdot Mw_X} \cdot 100\% = \frac{12 \frac{gC}{moleC} \cdot 1.05 moleC}{31.51 \frac{gX}{moleX} \cdot 1 moleX} \cdot 100\% = 39.99\%$$

$$N' = \frac{\delta \cdot Mw_N}{M \cdot Mw_X} \cdot 100\% = \frac{14 \frac{gN}{moleN} \cdot 0.32 moleN}{31.51 \frac{gX}{moleX} \cdot 1 moleX} \cdot 100\% = 14.22\%$$

$$H' = \frac{\beta \cdot Mw_H}{M \cdot Mw_X} \cdot 100\% = \frac{1 \frac{gH}{moleH} \cdot 1.83 moleH}{31.51 \frac{gX}{moleX} \cdot 1 moleX} \cdot 100\% = 5.81\%$$

Now we can place the substrate and bacteria parameters in the equation:

$$Y_{\frac{O}{X}} = \frac{32 \cdot C + 8 \cdot H - 16 \cdot O}{Y_X \cdot Mw_S} + 0.01 \cdot O' - 0.0267 \cdot C' + 0.01714 \cdot N' - 0.08 \cdot H'$$

$$Y_{\frac{O}{X}} = \frac{32 \cdot 3 + 8 \cdot 8 - 16 \cdot 3}{0.6 \frac{gX}{gS} \cdot 92 \frac{gS}{moleS}} + 0.01 \cdot 30.97 - 0.0267 \cdot 39.99 + 0.01714 \cdot 14.22 - 0.08 \cdot 5.81$$

$$Y_{\frac{O}{X}} = 1.05 \frac{gO_2}{gX}$$

We need to go back to the maximal oxygen consumption of the bacteria:

$$OUR_f = \mu \times X_f \times Y_{O_2/X} = 0.467 \frac{1}{hr} \times 20 \frac{gX}{L} \times 1.05 \frac{gO_2}{gX} = 10 \frac{gO_2}{L \times hr}$$

From the air flow rate, we can find the oxygen flow rate into the fermenter:

$$\begin{aligned} \frac{Q_{O_2}}{V_{ferm}} &= \frac{Q_{air}}{V_{ferm}} \times \%O_{2in} \times \frac{P}{T \times R} \times Mw_{O_2} \\ \frac{Q_{O_2}}{V_{ferm}} &= \frac{100 \frac{L_{air}}{hr}}{1L} \times 0.42 \frac{L_{O_2}}{L_{air}} \times \frac{1.5 atm}{295^\circ K \times 0.082 \frac{L_{O_2} \times atm}{K \times moleO_2}} \times 32 \frac{gO_2}{moleO_2} \\ \frac{Q_{O_2}}{V_{ferm}} &= 83.34 \frac{gO_2}{L \times hr} \end{aligned}$$

Therefore, in order to reach to a final concentration of $20 \frac{g \text{ biomass}}{lit}$, the minimal oxygen solubility in the fermenter should be:

$$\text{minimal solubility \%} = \frac{OUR}{\left(\frac{Q_{O_2}}{V_{ferm}}\right)_{in}} \times 100\% = \frac{10 \frac{gO_2}{L \times hr}}{83.34 \frac{gO_2}{L \times hr}} \times 100\% = 11.99\%$$

minimal solubility % = 12%

Question 3

A factory executes batch fermentation of bacteria. The fermenter volume is 1000 liter and at the beginning of the fermentation the fermenter is being inoculated with a starter contains 165 gr bacteria. Immediately after inoculation the logarithmic growth starts and lasts for 16 hours. When the growth is over the fermenter is harvested.

Normal air (21% oxygen) in a pressure of 1.5 atmospheres and temperature of 31°C is supplied to the fermenter in a constant flow rate of 2 VVM. The air exiting the fermenter is being monitored by system that continuously measures the % of oxygen in the air. At the beginning of the fermentation the measured % oxygen in the exiting air was 20.97%, while at the end of the fermentation the air exits with 17.53% oxygen.

A. What is the biomass yield on the oxygen ($Y_{X/O} \frac{\text{gr } X}{\text{gr } O_2}$)? (0.707 gX/gO₂)

B. What is the biomass concentration in the fermenter at the end of the fermentation (g/l)? (19.1 g/l)

Solution:

Given:

<u>Fermenter</u>	<u>Air</u>
$V_{ferm} = 1000L$	$P = 1.5atm$
$WX_0 = 165gX$	$T = 31^\circ C = 304^\circ K$
$t_{log} = 16hr$	$\left(\frac{Q_{air}}{V}\right)_{in} = 2 \frac{L_{air}}{L \times min}$
	$O_{2in} = 0.21 \frac{L_{O_2}}{L_{air}}$
	$O_{2out\ to} = 0.2097 \frac{L_{O_2}}{L_{air}}$
	$O_{2out\ tf} = 0.1753 \frac{L_{O_2}}{L_{air}}$

A. $Y_{\frac{X}{O}} = ?$

The biomass yield on the oxygen can be calculated from the oxygen uptake rate:

$$OUR = \frac{\mu \times X}{Y_{\frac{X}{O}}}$$

Assuming all the dissolved oxygen is consumed by the bacteria, we can find the OUR at the beginning and at the end of the fermentation by using the following:

$$OUR = \left(\frac{Q_{O_2}}{V_{ferm}}\right)_{in} - \left(\frac{Q_{O_2}}{V_{ferm}}\right)_{out}$$

We need to convert the unit from air flow to oxygen supply:

$$\frac{Q_{O_2}}{V_{ferm}} = \frac{Q_{air}}{V_{ferm}} \times \%O_2 \times \frac{P}{T \times R} \times Mw_{O_2}$$

Therefore:

$$\begin{aligned} \left(\frac{Q_{O_2}}{V_{ferm}} \right)_{in} &= 2 \frac{L_{air}}{L \times min} \times 0.21 \frac{L_{O_2}}{L_{air}} \times \frac{1.5atm}{304^\circ K \times 0.082 \frac{L_{O_2} \times atm}{K \times moleO_2}} \times 32 \frac{gO_2}{moleO_2} \times 60 \frac{min}{hr} \\ &= \mathbf{48.524 \frac{gO_2}{L \times hr}} \end{aligned}$$

$$\begin{aligned} \left(\frac{Q_{O_2}}{V_{ferm}} \right)_{out t_0} &= 2 \frac{L_{air}}{L \times min} \times 0.2097 \frac{L_{O_2}}{L_{air}} \times \frac{1.5atm}{304^\circ K \times 0.082 \frac{L_{O_2} \times atm}{K \times moleO_2}} \times 32 \frac{gO_2}{moleO_2} \times 60 \frac{min}{hr} \\ &= \mathbf{48.454 \frac{gO_2}{L \times hr}} \end{aligned}$$

$$\begin{aligned} \left(\frac{Q_{O_2}}{V_{ferm}} \right)_{out t_f} &= 2 \frac{L_{air}}{L \times min} \times 0.1753 \frac{L_{O_2}}{L_{air}} \times \frac{1.5atm}{304^\circ K \times 0.082 \frac{L_{O_2} \times atm}{K \times moleO_2}} \times 32 \frac{gO_2}{moleO_2} \times 60 \frac{min}{hr} \\ &= \mathbf{40.506 \frac{gO_2}{L \times hr}} \end{aligned}$$

We need to calculate the OUR at the beginning and at the end of the growth, while the oxygen uptake rate is the difference between the oxygen flow rate entering and exiting the fermenter:

$$\begin{aligned} OUR &= \left(\frac{Q_{O_2}}{V} \right)_{in} - \left(\frac{Q_{O_2}}{V} \right)_{out} \\ OUR_{t_0} &= (48.524 - 48.454) \frac{gO_2}{L \times hr} = 0.070 \frac{gO_2}{L \times hr} \\ OUR_{t_f} &= (48.524 - 40.506) \frac{gO_2}{L \times hr} = 8.018 \frac{gO_2}{L \times hr} \end{aligned}$$

Since $OUR = q_{O_2} \times X$, then $\frac{OUR_{t_f}}{OUR_{t_0}} = \frac{q_{O_2} \times X_f}{q_{O_2} \times X_0}$.

As the specific rate of oxygen uptake (q_{O_2}) does not change during this fermentation, the ratio between the air uptake at the end of the growth to the air uptake at the beginning of the growth is the ratio between the biomass concentrations:

$$\frac{OUR_{t_f}}{OUR_{t_0}} = \frac{X_f}{X_0} = \frac{8.018 \frac{gO_2}{L \times hr}}{0.070 \frac{gO_2}{L \times hr}}$$

$$\frac{X_f}{X_0} = 114.543$$

We will calculate the growth rate from the logarithmic growth rate:

$$\ln \frac{X_f}{X_0} = \mu \times t_{log} \rightarrow \mu = \frac{\ln \frac{X_f}{X_0}}{t_{log}} = \frac{\ln 114.543}{16hr} = 0.296 \frac{1}{hr}$$

The fermenter volume and the cells weight at the beginning of the fermentation are given, so we can calculate X_0 :

$$X_0 = \frac{W_{X_0}}{V_{ferm}} = \frac{165gX}{1000L} = 0.165 \frac{gX}{L}$$

Therefore, by using the uptake oxygen rate at the beginning of the fermentation. the biomass yield on the oxygen is:

$$Y_{\frac{X}{O}} = \frac{\mu \times X_0}{OUR_{t_0}} = \frac{0.296 \frac{1}{hr} \times 0.165 \frac{gX}{L}}{0.070 \frac{gO_2}{L \cdot hr}}$$

$$Y_{\frac{X}{O}} = 0.698 \frac{gX}{gO_2}$$

B. $X_f = ?$

In section A we found that:

$$\frac{X_f}{X_0} = 114.543$$

In section A we also calculated the biomass concentration at the beginning of the fermentation:

$$X_0 = 0.165 \frac{gX}{L}$$

Therefore, the biomass concentration in the fermenter at the end of the fermentation is:

$$X_f = 114.543 \times X_0 = 114.543 \times 0.165 \frac{gX}{L} = 18.899 \frac{gX}{L}$$

$$X_f = 18.9 \frac{gX}{L}$$

Question 4

A batch feeding is being used in a fermentation where it is not possible to feed the whole substrate amount in one time, as in the following case:

Lysine production by a certain bacterium that grows on glucose as the carbon source, is carried in batch fermentation in which the glucose concentration cannot go above 5%. The bacteria growth rate is 0.1 per hour and the Lysine production rate is $0.2 \frac{gr \text{ Lysine}}{gr \text{ Cell} \cdot hr}$. The initial bacteria concentration in the fermenter is 2 gr/lit and the initial glucose concentration is 5%. The aim of the fermentation is to reach final Lysine concentration of 50 gr per liter. It is known that when a similar bacterial strain, that was grown in similar conditions but for biomass production and not for Lysine production, the cells yield obtained on this substrate was 0.5 gX/gS.

A. Considering that substantial amount of the glucose in the medium is being consumed for Lysine production rather than for growth, what is the bacteria yield on glucose in this specific fermentation? Detail and explain every assumption made!

(0.167 gX/gS)

B. We feed a “pulse” of glucose to the fermenter whenever the last glucose amount is finished, in order to increase the glucose concentration back to 5%. What is the time schedule of the glucose additions until the end of the fermentation?

(It is required to add glucose at: 16.44 hr and 22.36 hr)

Solution:

Given:

<u>Bacteria</u>	<u>Fermentation</u>
$\mu = 0.1 \frac{1}{hr}$	$S_{0_{max}} = 5\% = 50 \frac{gS}{L}$
$q_P = 0.2 \frac{gP}{gX \times hr}$	$X_0 = 2 \frac{gX}{L}$
$Y_{\frac{X}{S}}^* = 0.5 \frac{gX}{gS}$	$P_f = 50 \frac{gP}{L}$
*Theoretical yield when Lysine is not produced.	

A. $Y_{\frac{X}{S}} = ?$

Assuming the literature data on $Y_{\frac{X}{S}}^*$ represents the maximal ability of the bacteria to utilize the substrate, we can conclude that:

$$Y_{\frac{X}{S}}^* = Y_{\frac{X}{S}} \times Y_{\frac{P}{S}}$$

We can calculate $Y_{\frac{P}{S}}$ by multiplying the growth yield by the production yield:

$$Y_{\frac{P}{S}} = Y_{\frac{X}{S}} \times Y_{\frac{P}{X}}$$

The production yield we can find from the specific production rate:

$$q_P = Y_{\frac{P}{X}} \times \mu$$

$$Y_{\frac{P}{X}} = \frac{0.2 \frac{gP}{gX \times hr}}{0.1 \frac{1}{hr}} = 2 \frac{gP}{gX}$$

Therefore, the Lysine yield on the substrate:

$$Y_{\frac{P}{S}} = Y_{\frac{X}{S}} \times Y_{\frac{P}{X}}$$

$$Y_{\frac{P}{S}} = Y_{\frac{X}{S}} \times 2 \frac{gP}{gX}$$

By placing in the original equation:

$$Y_{\frac{X}{S}}^* = Y_{\frac{X}{S}} + Y_{\frac{P}{S}}$$

$$Y_{\frac{X}{S}}^* = Y_{\frac{X}{S}} + 2 \times Y_{\frac{X}{S}}$$

$$Y_{\frac{X}{S}}^* = 3 \times Y_{\frac{X}{S}}$$

$Y_{\frac{X}{S}}^*$ is given in the question:

$$Y_{\frac{X}{S}} = \frac{0.5 \frac{gX}{gS}}{3}$$

$$Y_{\frac{X}{S}} = 0.167 \frac{gX}{gS}$$

Another explanation for a solution after finding the Lysine yield on the biomass:

From every 1 gram of biomass produced by glucose uptake, 2 gram Lysine are produced which also demand glucose. Meaning, in a fermentation that produce Lysine, there are 3 grams that need glucose, so the total amount of glucose in the process has to be divided between these 3 grams. If we assume that the carbon percentage in the microorganism is similar to the carbon percentage in the Lysine, we will get that on every gram glucose consumes, only third is used for growth purpose and the rest is for Lysine production. Therefore, in this fermentation, only 1/3 biomass is produced compared with a fermentation with no Lysine production, while the amount of consumed glucose is the same.

From here, in a fermentation with no Lysine production:

$$Y_{\frac{X}{S}}^* = \frac{\Delta X}{\Delta S}$$

In a fermentation with Lysine production:

$$Y_{\frac{X}{S}} = \frac{\frac{\Delta X}{3}}{\Delta S} = \frac{1}{3} \times \frac{\Delta X}{\Delta S}$$

By placing $Y_{\frac{X}{S}}^*$ in the equation instead of $\frac{\Delta X}{\Delta S}$ we get that:

$$Y_{\frac{X}{S}} = \frac{Y_X^*}{3}$$

$$Y_{\frac{X}{S}} = \frac{0.5 \frac{gX}{gS}}{3}$$

$$Y_{\frac{X}{S}} = 0.167 \frac{gX}{gS}$$

B. $\Delta t = ?$

We need to find the final cells concentration that will produce $50 \frac{gP}{L}$ product:

The production yield that was calculated in section A is bigger than 1, so we can assume that the product is extracellular and that $P_0 = 0 \frac{gP}{L}$. Therefore, the final cells concentration:

$$Y_{\frac{P}{X}} = \frac{\Delta P}{\Delta X} \rightarrow X_f = X_0 + \frac{\Delta P}{Y_{\frac{P}{X}}} = 2 \frac{gX}{L} + \frac{(50 - 0) \frac{gP}{L}}{2 \frac{gP}{gX}} = 27 \frac{gX}{L}$$

We need to calculate the biomass concentration at the point when the substrate is finished after every “pulse”, and from there we can find the fermentation period until that point:

$$Y_{\frac{X}{S}} = \frac{\Delta X}{\Delta S} \rightarrow X_2 = X_1 + Y_{\frac{X}{S}} \times \Delta S$$

$$\ln\left(\frac{X_2}{X_1}\right) = \mu \times \Delta t \rightarrow \Delta t = \frac{\ln\left(\frac{X_2}{X_1}\right)}{\mu}$$

$$X_1 = 2 \frac{gX}{L} + 0.167 \frac{gX}{gS} \times 50 \frac{gS}{L} = 10.35 \frac{gX}{L}$$

$$\Delta t_1 = \frac{\ln\left(\frac{10.35 \frac{gX}{L}}{2 \frac{gX}{L}}\right)}{0.1 \frac{1}{hr}} = 16.44hr = t_1$$

$$X_2 = 10.35 \frac{gX}{L} + 0.167 \frac{gX}{gS} \times 50 \frac{gS}{L} = 18.70 \frac{gX}{L}$$

$$X_3 = 18.70 \frac{gX}{L} + 0.167 \frac{gX}{gS} \times 50 \frac{gS}{L} = 27.05 \frac{gX}{L}$$

$$\Delta t_2 = \frac{\ln \left(\frac{18.70 \frac{gX}{L}}{10.35 \frac{gX}{L}} \right)}{0.1 \frac{1}{hr}} = 5.92 hr$$

$$t_2 = t_1 + \Delta t_2 = (16.44 + 5.92) hr$$

$$t_2 = \mathbf{22.36 hr}$$

No additional “pulse” is needed since the substrate will finish exactly when reaching to the biomass concentration required to produce $50 \frac{gP}{L}$.

To reach Lysine concentration of $50 \frac{gP}{L}$, we need to add glucose after 16.44 and 22.36 hr.